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Effects of annealing temperature on microstructure and mechanical properties of a low density δ-TRIP steel plate via intercritical quenching-annealing process

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ABSTRACT

In this study, the effects of annealing temperature on microstructure and mechanical properties of Fe–9Mn–5Al-0.3C (wt.%) low density steel plates via intercritical quenching-annealing process were investigated through field-emission scanning electron microscopy, electron backscatter diffraction, X-ray diffraction, and tensile tests. The results showed that the experimental steel maintained micro-laminated microstructure after intercritical quenching-annealing treatment, consisting of δ-ferrite band and tempered martensite lamellar. After annealing at the range of 600–800 °C following quenching of 1050 °C, tempered martensite lamellar mainly changed, and affected the mechanical properties of the steel. The annealed experimental steels exhibited wide yield strength grades (460–620 MPa) with excellent ductility (40–63 %) and low yield-tensile ratio (0.65–0.75). The proportion of austenite, the density of geometrically necessary dislocations, and high-angle grain boundary density of tempered martensite region, as well as the presence or absence of carbides, were considered to contributed to the mechanical properties. Austenitic mechanical ability was analyzed. Various annealed temperature led to the different levels of austenitic stability through changing the enrichment of alloyed elements in austenite, further causing different degrees of TRIP effect. TWIP effect existed during deformation according to estimation of SFE. The steel annealed at 740 °C had maximum uniform elongation (62.4 %) and the highest product of strength and elongation (45.2 GPa%), depending on the maximum proportion of austenite transformed into martensite during deformation. However, the 680 °C annealed steel with the highest austenitic stability exhibited deficient TRIP effect, leading to the occurrence of necking. Moreover, work hardening behavior and fracture morphology of the experimental steel were analyzed.

1. Introduction

In the recent years, Fe–Mn–Al–C steel aroused widespread attention in the fields of transportation metal material, offshore structure, and engineering machinery due to the social demand of lowering fuel consumption and carbon emission. Fe–Mn–Al–C steel was a promising type of steel with low density due to Al addition and high strength [1,2]. A 1.3 % reduction in density of steel could be obtained per 1 wt% addition of Al [3]. The Al addition promoted the form of δ-ferrite, but facilitated the nucleation of short-range order structure, which was harmful for ductility of metal. Therefore, Mn, as an austenite stabilization element, was usually added to steel together with Al. According to the content of Al and Mn, low density steel was classified into four categories: ferritic steels [4,5], ferritic-based duplex steels [6,7], austenite-based duplex steels [2,8], and austenite steels [9–11]. Given that low weldability and corrosion resistance, segregation, as well as higher cost, more than 12 wt % Mn was not recommended [12–14]. Hence, medium/low-Mn duplex low density steel was a suitable choice [15].

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Dual-phase low density steel composed of δ-ferrite and martensite/austenite was usually rolled and formed micro-laminated structure under deformation [16–19]. Below the homogenization temperature, it was difficult to refine δ-ferrite through recrystallization or phase transformation. Therefore, improving prior austenite area was an important method to enhance mechanical properties. Retained austenite could improve mechanical properties of steel by transformation induced plasticity (TRIP) and twinning induced plasticity (TWIP) [20–23]. The mechanical stability of retained austenite had been widely investigated [24,25]. Xu et al. [26] revealed the correlation between deformation behavior and austenite stability in Mn–Al type TRIP steel. Inhomogeneous stability of retained austenite could lead to inhomogeneous and discontinuous work hardening behavior, while the heterogeneity of austenitic grain size and the orientation of retained austenite played important roles in determining the stability of austenite when deformation. Chen et al. [27] reported that duplex lightweight steel of Fe–8Mn–6Al-0.4C exhibited excellent mechanical properties of tensile strength ~756 MPa and ~56 % elongation under synergy of TRIP and TWIP effects.

The different proportions of soft phase and hard phase in Micro-laminated low-density steel could lead to different mechanical properties. Shang et al. [28] changed the δ-ferrite content by various quenching temperature and tempering treatment to improve the mechanical properties of laminar low density steel. The total elongation of the steel was boosted from 4 % of hot-rolled state to 40 % of quenched-tempered state. Xu et al. [29] increased the retained austenitic content of micro-laminated low density steel by direction quenching and intercritical annealing, which improving the ductility form 13 %–41 % in total elongation under the assistance of TRIP effect.

In this study, the heat-treatment experiments of intercritical quenching (IQ) and annealing were carried out on the steel of Fe–9Mn–5Al-0.3C. The tensile strength, 0.2 % off-set yield strength, and total elongation of the experimental steel were investigated corresponding to the relevant laminated microstructure. The relationship between laminated microstructure and mechanical properties and the effects of the austenitic stability on deformation behavior were discussed. The main purpose of this research was to obtain better mechanical properties and heat-treatment parameters of the experimental steel to broaden its practical application.

2. Experiments and methods

The steel of Fe–9Mn–5Al-0.3C used in this study was prepared in a 5 kg ingot by a vacuum induction furnace. The actual chemical composition of the steel listed in Table 1. The density of the steel is 7.34 g*cm^{-3} measured by Drainage Method using Archimedes' Principle [30], and 6.7 % lower than that of low-carbon low-alloyed steels.

The ingot was heated to 1100 °C for 2 h, and further forged into a square forging ingot measuring 80 mm × 70 mm. Subsequently, after reheated to 1100 °C and held for 2 h, the forging ingot was hot rolled from 80 to 12 mm thickness in eight passes, with the initial rolling temperature approximately 1050 °C and the finishing rolling temperature approximately 950 °C, then air cooled to room temperature, and the corresponding sample labeled as HR. The hot-rolled experimental plate was reheated to 1050 °C, located in the ferrite and austenite dual-phase region, for 1 h, followed by water quenching, with the corresponding sample labeled as IQ-1050. The water-quenched plate was annealed for 1 h at various temperature of 620 °C, 680 °C, 740 °C, and 800 °C, with the corresponding samples labeled as QA-620, QA-680, QA-740, and QA-800. The schematic diagram of the thermal treatment of the experimental steel was shown in Fig. 1.

Table 1
the chemical compositions of the experimental steel (wt. %).

C	Al	Mn	P	S	Fe
0.29	4.6	8.9	< 0.005	< 0.005	Balance

Microstructure of the experimental steel was observed and analyzed by a field-emission scanning electron microscope (FE-SEM, MIRA-3 LMH) with an electron backscatter diffraction (EBSD, Oxford Instruments) probe. The metallographic specimens for SEM observation were polished by a polishing machine (PG-1A), and then etched by 10 % nitric acid alcohol. The specimens for EBSD observation and XRD were prepared by electrochemically polishing in electrolyte of 10 % perchloric acid, 5 % glycerol, and 85 % ethyl alcohol, with voltage of 15~20 V and electrical current of 0.8–1.2 A. HKL Channel5 software and Python were used to analyzed EBSD data. Thermo-Calc software with TCFE 12 database was used to calculate phase volume fraction diagram of the present steel, as Fig. 2 shown.

At room temperature, tensile properties of the steel under various treatment states were measured using the electronic tensile testing machine (WDW-100E) at the strain rate of 0.375 min^{-1}, with rod-shaped sample size of M10 * 60 mm and gauge length of 25 mm. The tensile axis was parallel to the rolling direction of the hot-rolled plate. X-ray diffraction (SmartLab) with Cu-Kα radiation at 40 kV and 150 mA was conducted to measure austenite content, with scanning step size of 0.02° and scanning speed of 2°/min. XRD specimens with scanning areas of 10 * 10 mm before tensile tests and 5 * 20 mm after tensile tests were prepared by the same method as EBSD specimens. Austenite volume fraction and estimation of stacking fault energy (SFE) were obtained by XRD Rietveld refinement analysis [31] using the FullProf software [32].

3. Results

3.1. Equilibrium phase diagram

Fig. 2(a) showed the phase fraction of the experimental steel under equilibrium state. Fig. 2(b) showed the mass fraction of C, Al, and Mn in ferrite, austenite as a function of temperature. Thanks to the addition of 4.6 % Al expanding δ-ferrite phase region, the experimental steel cannot be completely austenitized. Dual-phase zone with ferrite and austenite occupied a considerable range of temperature. δ-ferrite (δ-F, BCC_A2 in Fig. 2(a)) began to precipitate in the steel liquid (L) at around 1485 °C with the decrease of temperature. As long as temperature dropping to around 1390 °C, austenite (A, FCC_A1 in Fig. 2(a)) formed by the peritectic reaction, δ-F + L → A, with the solidification of the steel liquid. Subsequently, the volume fraction of austenite and δ-ferrite continuously increases and decreases, respectively. Meanwhile, the contents of Mn, C in ferrite decreased, whereas the content of Al increased. The content of Mn in austenite was relatively stable, while the content of C decreased and the content of Al increased. Until the temperature going down to around 970 °C, Austenite fraction reached to maximum, around 91.4 %, while δ-ferrite fraction reached to minimum, around 8.6 %. Considering the discrepancy of Thermo-Calc software, 1050 °C was selected as quenching temperature, making the volume of austenite

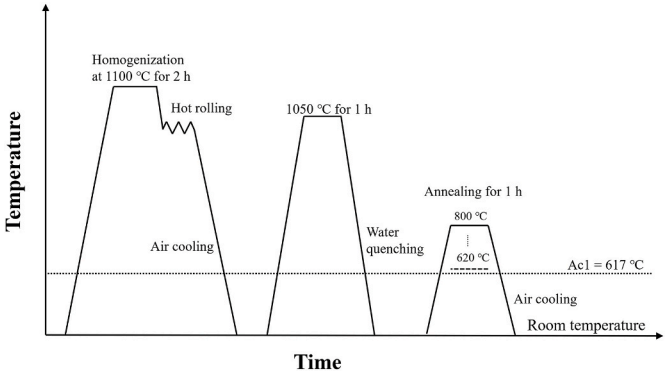

Fig. 1. Schematic diagram of the thermal treatment for the experimental steel.

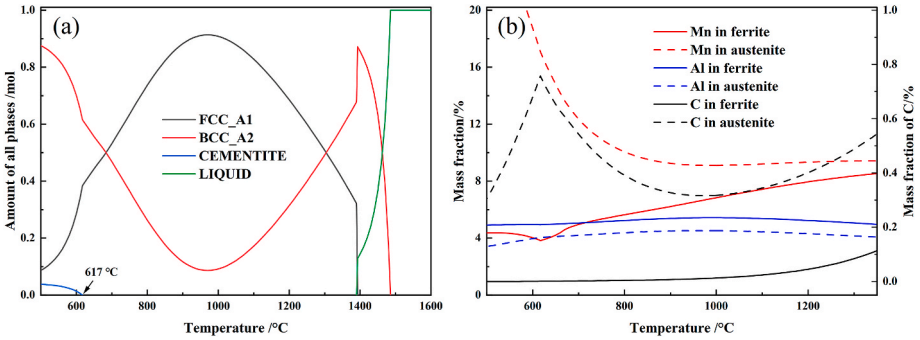

Fig. 2. (a) the phase fraction of the experimental steel calculated by Thermo-Calc software, and (b) the mass fraction of corresponding ferrite and austenite composition as a function of temperature.

become large enough. The volume fraction of austenite and δ-ferrite instead unceasingly decreased and increased, respectively, as the temperature dropping further. It is worthy noted that the contents of Mn, C in austenite intensively increased with the temperature decreasing from 970 °C. The content of C in austenite instead started to decrease when the temperature decreasing to around 617 °C as cementite precipitated. The A_1 temperature, the cementite precipitation point, was around 617 °C, and therefore 620 °C, 680 °C, 740 °C, 800 °C were selected as intercritical annealing temperature.

3.2. Microstructure of the experimental steel under various treatments

Fig. 3(a), (b) depicted SEM micrographs of the hot-rolled state (HR) of the experimental steel. A laminated microstructure consisting of δ-ferrite (δ-F) bands and lath martensite (M) was observed because of thermomechanical rolling in dual-phase region. Austenite underwent ferrite transformation and martensite transformation during the cooling process, and therefore α-ferrite (α-F) and lath martensite (M) formed. Moreover, retained austenite (RA) was observed in Martensite bands, which indicated the Mf temperature of the experimental steel was lower than room temperature.

The laminated microstructure was still existing in IQ-1050 due to its heredity [33], as shown in Fig. 3(c). Whereas, prior austenite grain grew up as lath martensite reversed in the holding process at 1050 °C, and prior austenite grain boundary (PAGB) moved to δ-ferrite. Accordingly, the δ-ferrite bands became thinner, and even were divided into some discontinuous δ-ferrite grains by prior austenite, as shown in Fig. 3(d). High heating temperature of 1050 °C made the Mn content of reversed austenite become relatively low according to Fig. 2(b), decreasing the stability of austenite and increasing Ms temperature. Further, the higher cooling rate of water quenching than air cooling inhibited α-ferrite transformation and increased Ms temperature as well [33]. Therefore, martensite region of IQ-1050 was larger and retained austenite was fewer at room temperature.

After annealing process, lath martensite area changed obviously. Fig. 3(e)~(h) showed the SEM micrographs of tempered martensite in QA specimens. Carbides (Cm), which were considered to be cementite, were precipitated and widely distributed in tempered martensite, only when annealing temperature was 620 °C, as shown in Fig. 3(e). Fine, reverted austenite grains were observed among the tempered martensitic laths of the four annealing temperatures. Furthermore, the new martensite from austenite by martensite transformation existed in the tempered martensite region of QA-800, as shown in the red box of Fig. 3 (h). As the annealing temperature increasing to 800 °C, small austenite grains grew up, and C, Mn elements in the austenite were diluted. Therefore, Ms temperature of austenite increased, resulting in Martensite transformation during air cooling.

Band contrast (BC) maps with clearer austenitic distribution of the various specimens were presented at Fig. 4. The brighter and whiter regions due to high BC value were consider as ferrite, while darker regions due to low BC value were consider as lath martensite [34,35], as shown in Fig. 4(a and b). Tempered martensite in QA series possessed relative higher BC value than martensite, which was related to recovery of martensite in annealing treatment leading to decreasing of dislocations. According to Fig. 4(a, b), δ-ferrite and prior austenite grains grew up after heating of 1050 °C, and ferrite bands were segmented by martensite lamellar, becoming dispersed. This situation was kept in annealed microstructure. For annealed specimens, fine austenite grains, including retained austenite after quenching treatment and reverted austenite, were observed to be widely distributed in tempered martensite region, as shown in Fig. 4(c, d, e, f) the distribution of austenite in HR was more concentrated in comparison with QA specimens, while there was little austenite in IQ-1050, as shown in Fig. 4(a and b). Table 2 listed the austenitic volume fractions of QA specimens measured by EBSD and XRD. Although the difference of the values obtained by the two methods existed, the changing of them with annealing temperature showed a similar trend. With increase of annealing temperature at the range of 620–740 °C, austenitic proportion increased. However, the difference of austenitic volume fraction between QA-740 and QA-800 was unobvious, and even the fraction of QA-800 was lower, which could be associated with martensite transformation of the latter when air cooling. In addition, the austenitic volume fraction of HR and IQ-1050 measured by EBSD, listed in Table 2 as well, was lower than QA specimens. The following discussion about austenitic volume fraction of QA specimens was based on the XRD data.

3.3. Mechanical properties

Fig. 5(a) and Table 3 showed engineering stress-strain curves and specific mechanical performance indicators of the experimental steels at room temperature, respectively. The hot-rolled steel showed the highest tensile strength (TS) of ~1107 MPa and poor total elongation (TE) of ~5.6 %. After intercritical quenching, the yield strength (YS) of the steel was improved, ~575 MPa, compared with HR, but TS was declined, ~929 MPa, and TE of ~2.0 % was poorest among all of specimens.

Annealing process following quenching considerably enhanced total elongation of the steel. With the annealing temperature increasing, TEs of QA series steels were ~44 %, ~45.6 %, ~62.4 %, and ~50.4 %, respectively. The TEs increased as annealing temperature until annealing temperature was 800 °C, which was relevant to martensite existing in QA-800. Moreover, all tensile specimens occurred different degrees of uniform deformation, which associated with special deformation mechanism of a large number of austenite, such as TRIP effect. Uniform elongation of all QA specimens, except QA-680, was almost equal to total elongation. For QA-680, When strain reached to around 25 %, the stress began to decrease, which meant that there was significantly necking in the tensile specimen. With the increase of annealing temperature, both of YS and TS of QA specimens were gradually decreased. It was worth noting that the YS and TS of QA-620 was much higher than that of other QA specimens, which was principally associated with

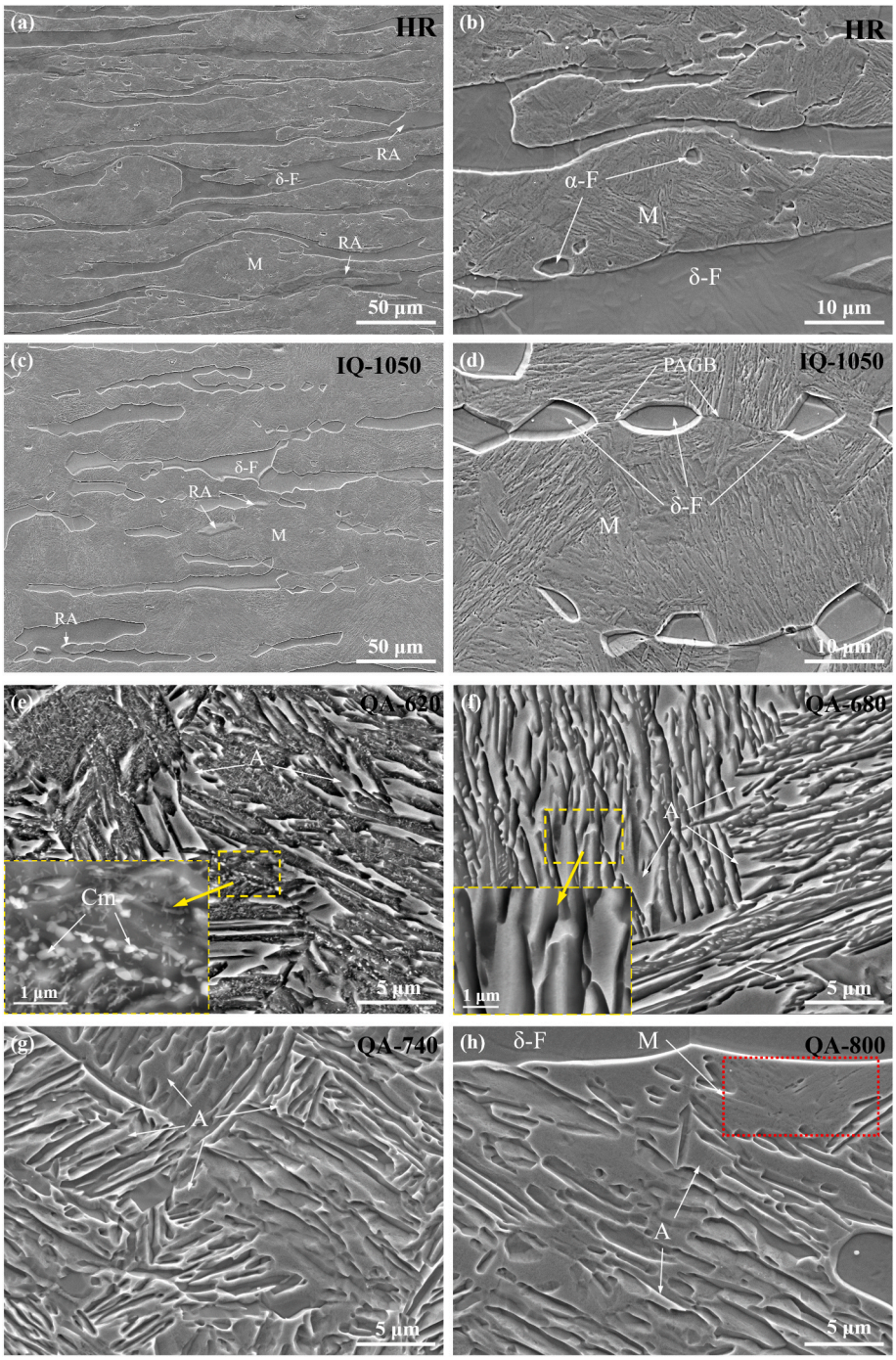

Fig. 3. SEM micrographs of the experimental steels with various states, where M referring to martensite, RA referring to retained austenite, δ-F referring to δ ferrite, α-F referring to α ferrite, PAGB referring to prior austenitic grain boundary, A referring to austenite, and Cm referring to carbides.

precipitated carbides in QA-620. The products of tensile strength and total elongation (PSE) of all QA specimens were more than 30 GPa%, showing superior comprehensive mechanical properties. PSE of QA-740, ~45.2 GPa%, was the highest one, which means that QA-740 owns the most excellent combination of strength and ductility. Furthermore, all QA specimens exhibited low yield-to-tensile ratio. Overall, the experimental steel after QA treatment exhibited wide yield strength grades (460–620 MPa) under the conditions of excellent ductility and low yield-to-tensile ratio, and was a typical third generation advanced high strength steel. Relying on the advantage of low density, the experimental steel could be used for high strength automobile steel plates, substituted for TRIP 590T/400Y from SAE J2745-2007 standard [36].

Work hardening rate (WHR) curves were plotted in Fig. 5(b). At the initial stage of $\varepsilon < 0.01$, the WHR sharply decreases, which was mainly related to deformation of soft phase, for example, ferrite, and dislocation movement [37,38]. For $\varepsilon > 0.01$, WHR curves got used to slightly decrease, which was attributed to hardening mechanism. WHR curves of all QA specimens were almost superimposed at the range of $\varepsilon < 0.15$. However, for true strain of 0.01–0.1, the WHR of QA-620 was apparently higher than that of other QA specimens, which could be attributed to precipitated carbides intensely retarding dislocation gliding. Since true strain was over 0.15, WHR of QA specimens had been shown different trends, which was relevant to the various levels of TRIP effect.

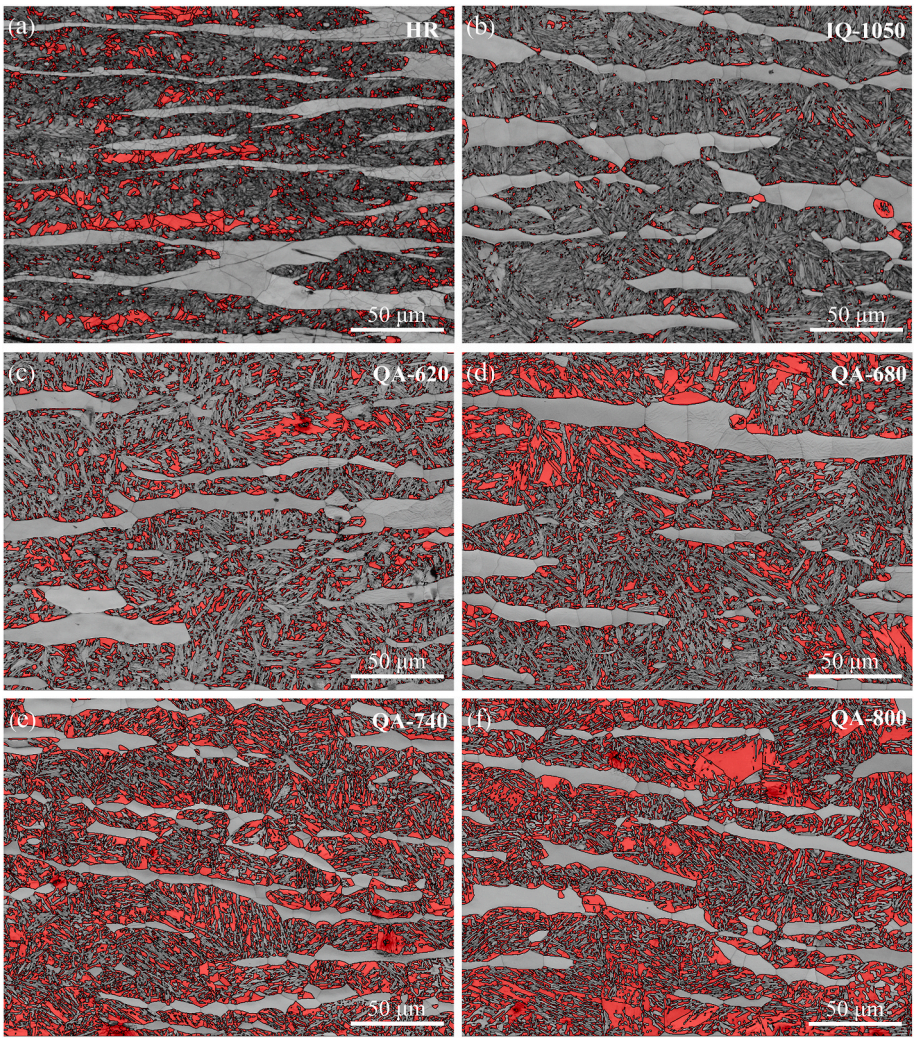

Fig. 4. Austenite (red) distribution and BC maps of the experimental steels under various treatments. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2
Austenite volume fraction (%) of the experimental steel measured by EBSD and XRD.

	HR	IQ-1050	QA-620	QA-680	QA-740	QA-800
By EBSD	~16	~7	~28	~35	~50	~49
By XRD	/	/	~33	~37	~51	~48

3.4. Fracture characterization

As shown in Fig. 6 was SEM morphology of tensile fracture of QA specimens, which all exhibited mixed fracture feature with cleavage in the red dashed areas, quasi-cleavage in the blue dashed areas, and dimples in the yellow dashed areas. Second cracks were observed in these fractures as well due to stress concentration in interface between δ and γ/α. In the cleavage areas, which associated with δ-ferrite [39,40], some cleavage features, including river pattern and cleavage step, were

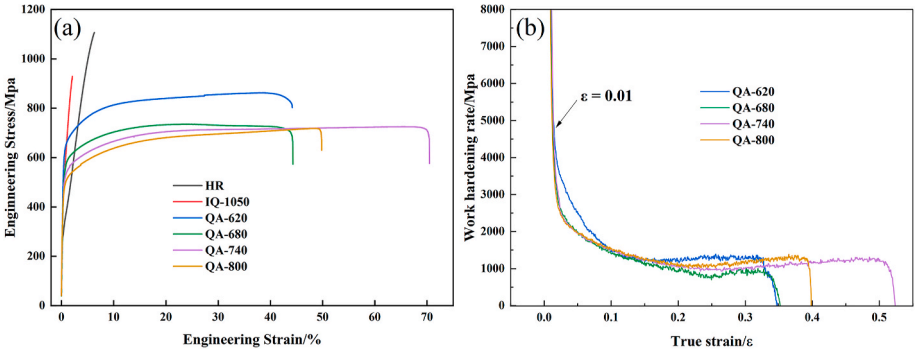

Fig. 5. (a) Engineering stress-strain curves of HR, IQ-1050, and QA specimens, and (b) work hardening rate curves of QA specimens.

Table 3
Mechanical properties of the specimens with various treatment states.

Specimens	YS/MPa	TS/MPa	TE/%	Yield-Tensile ratio	PSE/GPa %
HR	305 ± 11	1107 ± 32	5.6 ± 0.4	0.28	6.2 ± 0.6
IQ-1050	575 ± 13	929 ± 30	2.0 ± 0.3	0.62	1.9 ± 0.4
QA-620	614 ± 2	864 ± 15	44.0 ± 3.5	0.71	38.0 ± 4.0
QA-680	542 ± 3	735 ± 12	45.6 ± 2.7	0.74	33.5 ± 2.5
QA-740	498 ± 3	725 ± 9	62.4 ± 4.2	0.69	45.2 ± 3.7
QA-800	469 ± 4	719 ± 10	50.4 ± 3.2	0.65	36.2 ± 2.9

presented, and indicated by white arrows with corresponding label. Due to the laminar microstructure with δ-ferrite band and tempered martensite, laminar tears indicated by red arrows were observed [28]. For QA-620 shown in Fig. 6(a), significantly smaller micro-dimples than other specimens were observed. River pattern displayed the propagation of cleavage cracks, which were stopped by toughness fracture areas, dimples. For QA-680 and QA-740 shown in Fig. 6(b and c), cleavage/quasi-cleavage and dimples areas have not sensible borders, and instead crisscrossed together. Furthermore, larger dimples pointed by yellow arrows existed in the fractures than normal agglomerated dimples in the yellow dashed areas. In Fig. 6(d) of QA-800, cleavage layer showed a mainly feature of cleavage step, which alternated with dimples layer.

4. Discussion

4.1. Microstructure evolution and mechanical properties

Mechanical properties of steels are intimately correlated with the microstructure. Adding Al and Mn elements promoted the formation of a dual-phase layered microstructure in the steel used in this research, consisting of δ-ferrite and austenite. Upon cooling to room temperature, the austenite, due to its weak stability, underwent martensitic transformation. The variations in prior austenite lamellar in various treatments exerted important influence on mechanical properties of the present steels. Fig. 7 showed band contrast maps, focusing on prior austenite lamellar, of the present steels under various processes, and corresponding kernel average misorientation (KAM) maps and grain boundary distribution maps. geometrically necessary dislocations (GND) densities of corresponding specimens were obtained by following equation [41,42]:

$$\rho^{GND} = 2KAM_{ave}/\mu b \tag{1}$$

$$KAM_{ave} = \exp\left(\frac{1}{N}\sum_{1}^{i} \ln KAM_i\right) \tag{2}$$

Where, ρ^{GND} was the GND density of calculated area, KAM_{ave} was the average local misorientation of the calculated area, b was the Burger's vector (2.887 Å for BCC, 2.563 Å for FCC, measured by XRD), μ was the step sizes (150 nm) of EBSD data, KAM_i was the local KAM value at pixel point i, and N was the number of pixel points of the calculated area. The calculated values of GND for BCC phase and FCC phase were shown in Fig. 8(a).

The microstructure of HR subjected to hot-rolling conditions showed the highest GND density of BCC phases and FCC phase. For BCC phases including ferrite and martensite, relatively more ferrite in HR, as shown in Fig. 7(a), led to higher real GND density of lath martensite, which gave the steel high tensile strength (1107 ± 32 MPa) but poor ductility (5.6 % ± 0.4). During quenching processing of 1050 °C, all martensite in the investigated hot-rolled steel reversed into austenite, and the diffusion of C and Mn elements to austenite led to more austenitic volume fraction, further resulting in more lath martensitic fraction at room temperature. Heating of up to 1050 °C could largely decrease GND density of hot-rolled microstructure, but considerable martensitic transformation during quenching would lead to volume expansion,

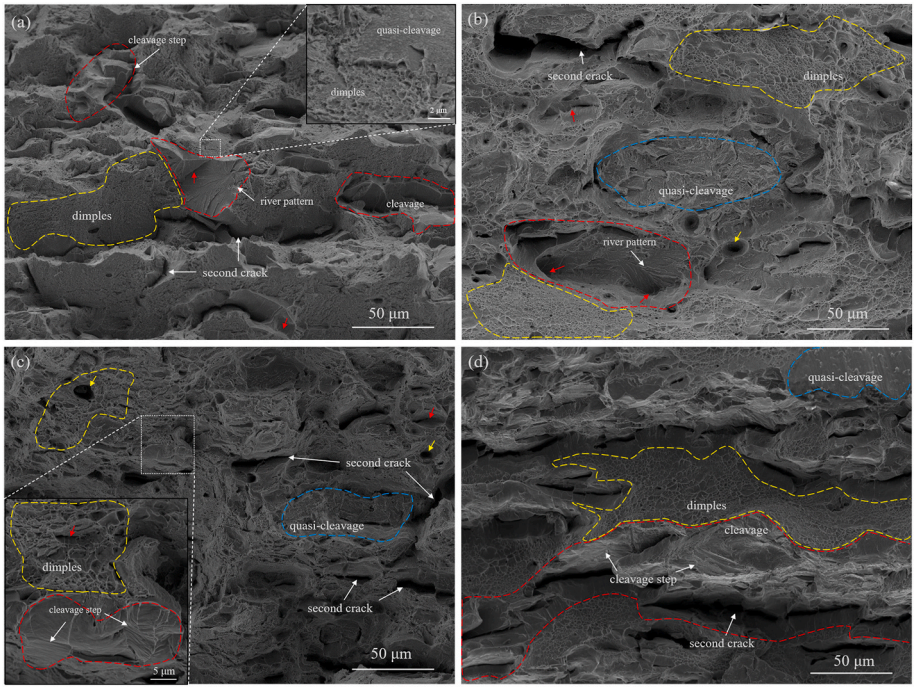

Fig. 6. SEM micro-morphology of tensile fracture of (a) QA-620, (b) QA-680, (c) QA-740, and (d) QA-800, while red dashed areas are cleavage, blue dashed areas are quasi-cleavage, and yellow dashed areas are small dimples, with red arrows pointing at laminar tears, and yellow arrows pointing at large dimples. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

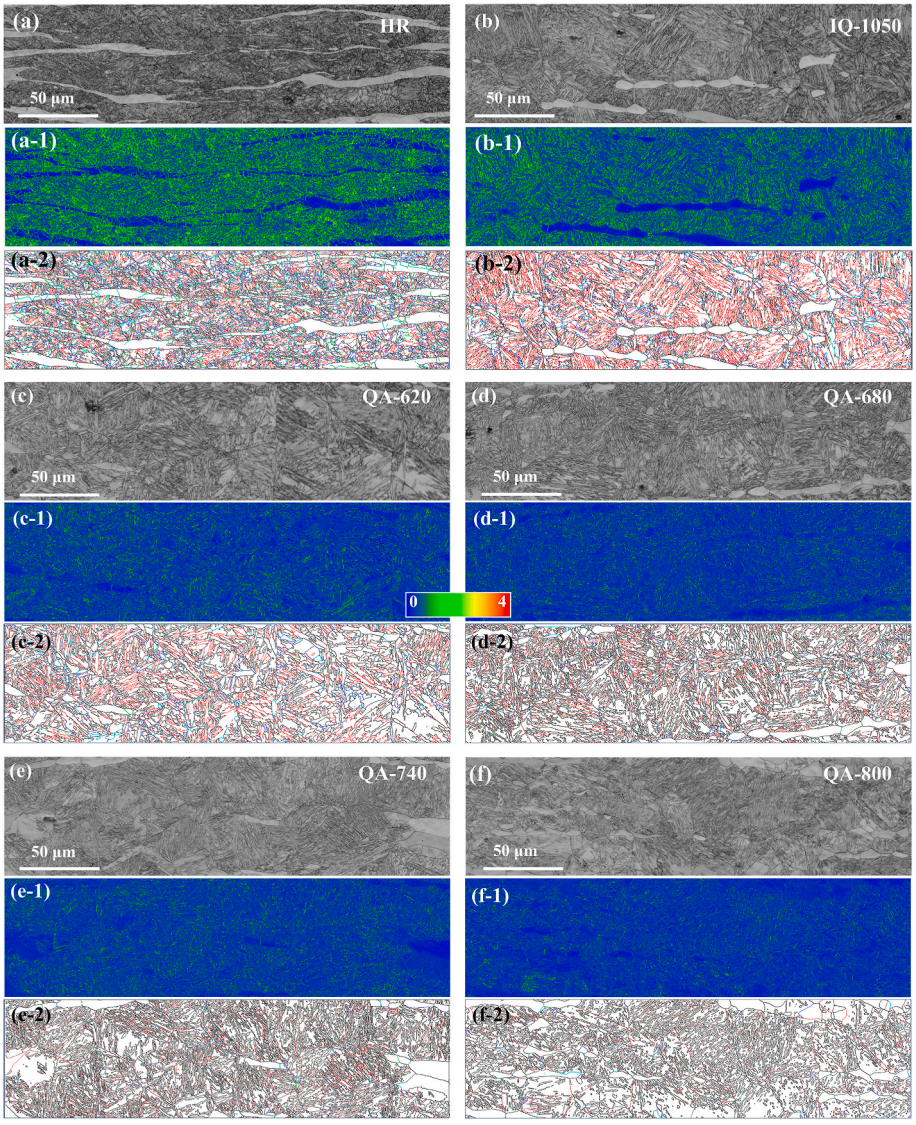

Fig. 7. (a, b, c, d, e, f) Band contrast (BC) maps of QA specimens focusing on tempered martensite, and (a-1,b-1,c-1,d-1,e−1,f-1) corresponding kernel average misorientation (KAM) maps, as well as (a-2,b-2,c-2,d-2,e−2,f-2) grain boundary distribution maps with black lines of AG boundaries, blue lines of high-angle packet boundaries, cyan lines of low-angle packet boundaries, red lines of block boundaries, and green lines of sub-block boundaries. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

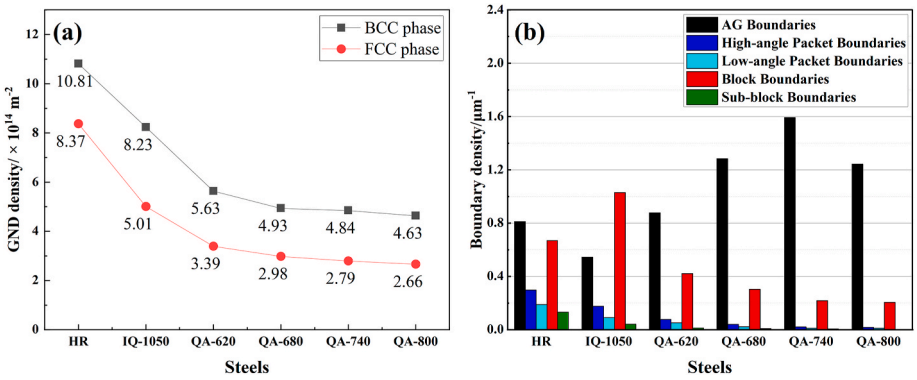

Fig. 8. (a) GND density and (b) grain boundary density of the experimental steel corresponding to Fig. 7.

which still resulted in high GND densities of lath martensite and retained austenite. More lath martensite in IQ-1050 than HR contributed to higher yield strength (575 ± 13 MPa), but more easily induced stress concentration during deformation causing premature failure. Therefore, the IQ-1050 had a poorer tensile strength (929 ± 30 MPa) and ductility (2.0 % ± 0.3). Furthermore, austenite of HR and IQ-1050 had similar

GND density values compared to that reported for strained Hadfield austenitic steel [43,44] under true strain of ~0.05, whose GND density value was around 9.80 × 10^{14}/m^2.

Annealing process following IQ process considerably improved the comprehensive mechanical properties of the steel in this study, but decreased the yield strength of QA specimens. On the one hand, during annealing processing, retained austenite fraction after quenching increased, while reverted austenite nucleated and grew up at martensite lath gaps, with the diffusion of the carbon and manganese into austenite grains, and aluminum into martensite and ferrite. It was widely accepted that ductile deformation started in soft phase universally, such as ferrite, austenite [45]. With increase of annealing temperature, the fraction of austenite gradually increased until QA-800 (Table 2). That the soft phase of austenite increasing will be prone to induce ductile deformation, leading to yield strength decreasing. On the other hand, recovery effect of annealing treatment led to the decrease of the GND density in tempered martensite and austenite. With the increase of annealing temperature, the martensitic and austenitic GND values of annealed specimens decreased continuously. The decrease of the GND density will be easier to cause dislocation slip during deformation, decreasing the yield strength of tempered martensite lamellar. Low GND density in tempered martensite promoted the coordinate deformation between tempered martensite and δ-ferrite band, which was beneficial to plasticity of the steel [28].

Therefore, the fraction of austenite, and the GND density of tempered martensite lamellar are the mainly reasons of decreasing yield strength of the experimental steel by annealing process. Furthermore, for QA-620, the precipitation of carbides strongly retarded dislocation gliding to greatly enhance yield and tensile strength, while the decrease of alloy elements content in matrix reduced the lattice distortion of tempered martensite to rise the deformability of the steel, which was positive to the ductility. For QA-800, though its austenite fraction was slightly lower than that of QA-740, its GND densities of tempered martensite and austenite was lower, still leading to lower YS.

Grain boundaries of alloy materials plays a significantly undeniable role in mechanical properties. Depending on the angle of boundaries, grain boundaries are generally classified into three ranges: 5°-15°, 15°-45°, and >45° [28,46], where high angle boundaries (>15°) are regarded as critical to affect mechanical properties [47]. A Python machine learning model developed by Li et al. [48] based on crystallographic information were used to correctly identify the various boundary types and exactly calculate the boundary density, as shown in Fig. 7(a-2,b-2, c-2,d-2,e−2,f-2). Given that the coarse δ-ferrite grains, the grain boundaries mainly gathered in martensite layer. Hence, grain boundaries can affect the mechanical properties of the present steel only by affecting the martensitic properties. As is seen in Fig. 8(b), austenitic grain (AG) boundaries (15°-45°), including austenitic boundaries and prior austenitic boundaries, and block boundaries (>45°) in all specimens were the highest two of boundary density. After quenching of 1050 °C, the density of block boundary was higher than that of AG boundary, and the situation of HR was opposite, which indicated that more austenitic fraction underwent martensitic transformation than that of HR. the density of block boundary increased also contributed to the yield strength of IQ-1050. For QA specimens, with the increase of annealed temperature, AG boundary density increased. When annealed temperature being 800 °C, the density instead decreased due to some coarse austenite grains. Meanwhile, block boundary density decreased continuously with annealed temperature. With the increase of annealed temperature, increasingly reverted austenite nucleated and grew up at the martensitic lath gaps, which resulting in that block boundaries gradually replaced by AG boundaries. Obviously, the variation trend of AG boundary density was consistent with that of austenitic volume fraction, but it did not play an effectively role in enhancing strength. Luo et al. [49] had reported that block boundaries remarkably contributed to yield strength of martensitic stainless steel. It was considered that the decrease of block boundary with angle of >45° in density, to some extent, contributed to the decrease of yield strength of the annealed specimens, with the increase of annealed temperature [50].

4.2. The austenitic stability

Superior ductility of medium-Mn low density steel depended on some special deformation mechanisms, such as TRIP and TWIP effects, which were in turn closely associated with SFE of austenite reflecting austenitic mechanical stability. In general, TRIP mechanism occurred in steel with SFE <20 mJ/m^2, while TWIP mechanism occurred in steel with SFE between 20 mJ/m^2 and 40 mJ/m^2. To estimate the austenitic SFE of the experimental steel, equation (3) was adopted [51]:

$$SFE = \frac{6.6 G_{(111)} a_0 A^{-0.37}}{\sqrt{3}\pi} \frac{\varepsilon_{50}^2}{SFP} \tag{3}$$

Where, A was the Zener elastic anisotropy, $G_{<111>}$ was the shear modulus in <111> direction, a_0 was the lattice constant, ε_{50}^2 was root mean square microstrain in the <111> direction averaged over the distance of 50 Å, and SFP was stacking fault probability. These parameters were obtained by Rietveld refinement analysis for XRD profiles (Fig. 9(a)) [31], or gained in the reference [51]. The estimated SFEs of austenite were listed in Table 4. The SFEs of QA specimens were between 20 mJ/m^2 and 40 mJ/m^2, which meant that TWIP effect was supposed to exist in austenite when deformation. The SFEs of QA specimens changed with annealing temperature, which was largely related to the variation of chemical composition in austenite [52]. For QA-620, precipitation of carbides and lower annealed temperature reduced the SFE of the austenite in comparison with QA-680. Above the annealing temperature of 680 °C, SFE decreased with the temperature increasing due to elemental partitioning.

The aforementioned content estimated SFEs of the experimental steel to assess the mechanical stability of austenite only based on XRD data before tensile test. The following content would more deeply analyze the deformation mechanism of QA specimens and the mechanical stability of austenite, based on experimental data before and after tensile tests. XRD profiles (Fig. 9(a)) of QA specimens before and after tensile deformation were analyzed for corresponding austenitic volume fraction, as shown in Fig. 9(b). It was concluded that TRIP effect, in addition to TWIP effect, also occurred during deformation, although the SFEs >20 mJ/m^2, because austenitic fraction after tensile test was lower than that before tensile test. To quantitively analyzed austenitic stability, equation (4) was adopted [53,54]:

$$V_\varepsilon = V_o \exp(-k\varepsilon) \tag{4}$$

Where, V_o and V_ε represented the austenitic volume fraction before and after tensile tests, respectively, ε represented corresponding true strain after tensile tests, and k represented the austenitic mechanical stability of various specimens. The higher k value meant lower austenitic stability, as depicted in Fig. 9(c). Therefore, QA-740 and QA-800 with the highest two k values, ~1.47 and ~1.50, exhibited more notable TRIP effect, while QA-680 with the lowest k value, ~0.56, showed little austenite to martensite transformation. Moreover, the estimated SFE of QA-680 was also the highest. With the annealing temperature increased, the variations of k value and SFEs display a opposite trend (Fig. 9(c)). The reflections of the two indictors on austenitic mechanical stability was instead consistent. This was that the lower the k value was, the higher the SFE was, which meant the higher mechanical stability of austenite. The stability of austenite was related to the enrichment of alloy elements, temperature, and grain size [52,55,56]. For QA-620, the precipitation of carbides led to a decrease of alloy elements in austenite, which diminished its austenitic stability, compared to QA-680. As annealed temperature increased, the dislocation nodes decreased, which also led to an increase of austenitic SFE, i.e., austenitic stability in QA-680, compared to QA-620. Above 680 °C, austenite fraction increased with the increasing of annealing temperature, which resulted

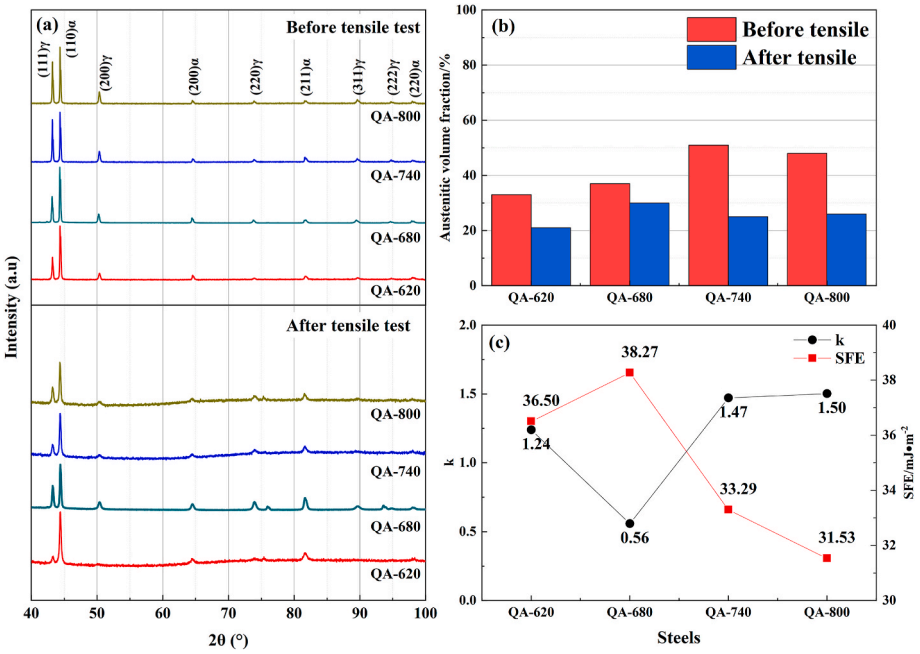

Fig. 9. (a) XRD profiles of QA specimens before and after tensile test, and (b) austenitic volume fraction bar chart of QA specimens, as well as (c) corresponding dot-line charts of *k*, austenitic stability constant, and SFE.

Table 4
Calculated SFE of QA specimens and some parameters.

Specimens	SFP/ $\times 10^{-4}$	ε_{50}^{2}/ $\times 10^{-5}$	SFE/mJ·m^{-2}
QA-620	~5.83	~11.68	~36.50
QA-680	~5.98	~12.33	~38.27
QA-740	~5.89	~10.76	~33.29
QA-800	~5.59	~9.68	~31.53

in that the alloying elements in austenite were diluted. Thus, austenitic stability decreased with annealing temperature. When annealing temperature was 800 °C, a part of extremely unstable austenite grains had undergone martensitic transformation in advance during air cooling of annealing, causing V_o, austenitic fraction before tensile test, to be smaller. There was relatively lower austenite transformed into martensite in QA-800 during deformation than QA-740. Therefore, calculated *k* value of QA-800 was not much higher than that of QA-740. Furthermore, during annealing treatment, reverted austenite grains nucleated and grew up in tempered martensite. There were some reverted austenite grains with various grain sizes. Smaller grains had higher enrichment of alloy elements, and their stability was higher, while larger austenite grains possessed lower stability.

Apparently, the outstanding ductility of QA specimens was inevitably traced back to TRIP effect. More than half of the austenite in QA-740 was participated in TRIP effect, causing the best total elongation. Due to more austenite underwent TRIP effect in QA-620 than that of QA-680, which neutralized the improvement of ductility resulting from the lower GND density due to higher annealed temperature of the latter, the total elongation of the former was almost the same as that of the latter. For QA-800, not only aforementioned, newly formed martensite during air cooling of annealed treatment deteriorated ductility, less austenite experiencing TRIP effect than QA-740, but also, produced relatively lower ductility. Moreover, when some austenite transformed into martensite during deformation, some of the rest of austenite should produce TWIP effect [27].

4.3. Deformation behavior

The complete process of deformation was analyzed here. Fig. 10 depicted separately the work hardening rate (WHR) curves and true stress-strain curves of the QA specimens. The WHR curves of all QA specimens had an analogous shape, and could be divided into three stages based on the slopes of WHR curves. In stage I, dislocation gliding started sequentially in ferrite and austenite, as well as tempered martensite, according to the difference of phase in hardness [29]. Particularly, after heating process of 1050 °C and annealed treatment, δ-ferrite grain apparently grew up, while small grains were gradually engulfed by larger grains. Hence, the grain boundary density of ferrite apparently decreased, while dislocation mean free path increased. Derived from large dislocation mean free path of ferrite, the WHR sharply decreased [57,58]. In stage II, the decline of the WHR obviously slowed down, which attributed to several hardening mechanisms. The hindrance of grain boundaries and high-density dislocation tangles to dislocation movement, as well as TRIP effect, could enhance WHR. Meanwhile, dislocation gliding played a mainly role in reducing WHR. Moreover, the finite dislocation annihilation at grain boundaries [59], as well as cross slip among dislocations [60], were also expected to contribute to lowering WHR. Apparently, dislocation gliding was still dominant as WHR slowly decreased in stage II.

In stage III, the WHR instead increased slowly, which meant TRIP effect dominating. Due to finer austenite grains after annealing process, the effect of austenitic grain size on TRIP effect was non-negligible. Fig. 11 showed austenitic grain size statistics of QA specimens before deformation. The distributions of austenite grain size in all QA specimens had a similar trend. With the decrease of grain size, the quantities of austenite grain gradually increased, and the size of around 90 % of the grains was less than 4.8 μm. The finer austenite grain possessed higher enrichment degree of alloy elements, and thus corresponding austenitic stability was higher [61]. The fine austenite grains possessing different stability transformed into martensite only when a certain critical strain was attained [62]. Therefore, in stage II, a small number of austenite grains with relatively large size underwent TRIP effect, producing weak hardening effect on WHR. As true strain increasing during stage III, a larger number of small austenite grains transformed into martensite,

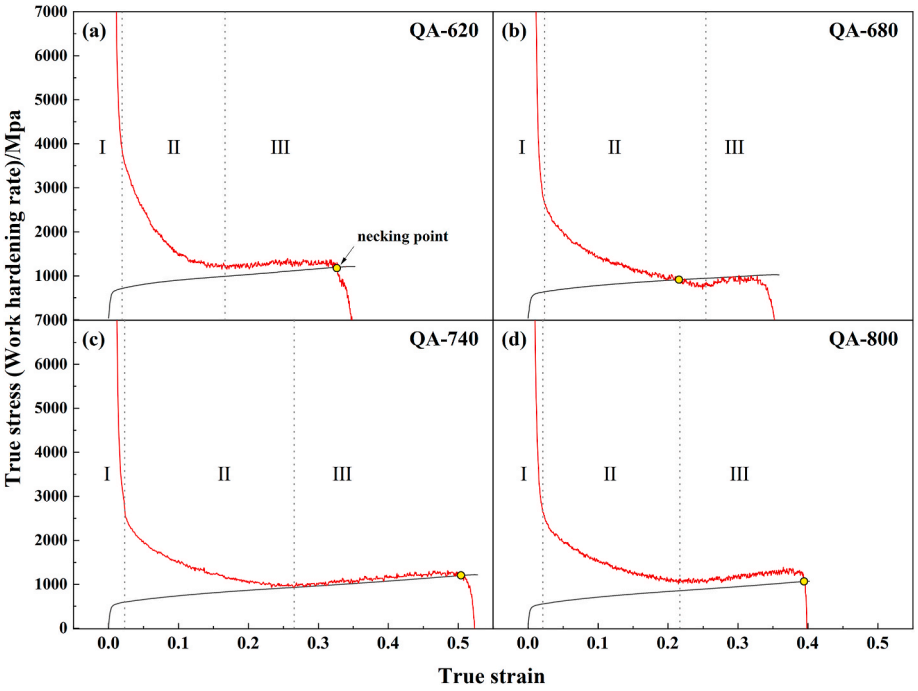

Fig. 10. Work hardening rate curves (red lines) and true stress-strain curves (black lines) of QA specimens, with the yellow point being necking point [66,67]. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

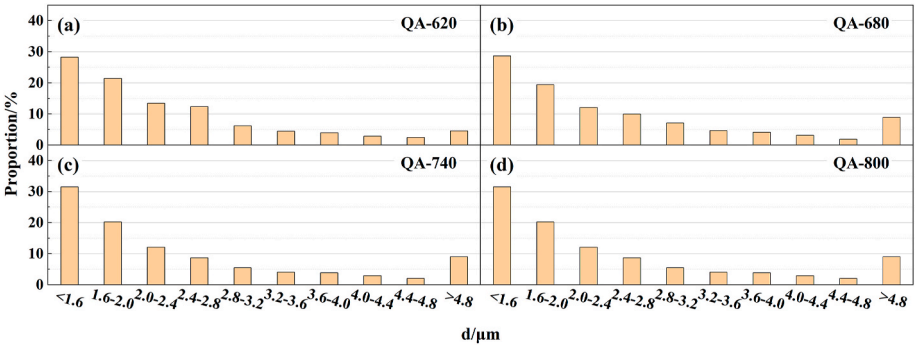

Fig. 11. Austenitic grain size statistics of QA specimens before tensile tests from EBSD data, with the diameter, d, obtained by equivalent circle method.

producing stronger hardening effect to make WHR enduringly increasing. The martensite produced during the strain process not only acted as an obstacle of dislocation gliding, but also as a site of dislocation accumulation and a source of partial dislocations [7]. Further, the martensite due to fine austenite grains, as small particles of hard phase, distributed widely in tempered martensite area, which could hinder dislocation gliding more strongly, boosting WHR. This was similar to second-phase particles strengthening mechanism. Moreover, based on the estimation of SFE (section 4.2), TWIP effect could occur when deformation. TWIP effect might play a role in the later stage of stage III [27].

Moreover, according to the necking points, all QA specimens, except for QA-680, completely experienced uniform deformation during tensile tests. For QA-680, with the necking occurring, the critical strain required for martensitic transformation was still obtained, and metastable austenite produced TRIP effect which instead enhanced WHR. Therefore, QA-680 exhibited another uniform elongation after necking, instead of direct failure. TRIP effect played a determining role in the occurrence of necking points.

During tensile tests, ductility deformation led to dislocation accumulation at the interfaces of δ-ferrite band and tempered martensite lamellar. Due to the difference in hardness between the two phases [29], stress concentration was prone to occur at their interfaces, to induce crack initiation. Once the crack forms, it would be captured by δ-ferrite, and propagated into and along δ-ferrite band, leading to the formation of second cracks like deep valleys (Fig. 6) [39,63]. Moreover, the crack propagating into δ-ferrite could be blunted due to the soft phase of δ-ferrite, which produced high resistance to the cracking [63]. Strain-induced martensitic transformation led to volume expansion, promoting the closure of cracks [64], which was positive to the ductility. Deficient high-angle grain boundaries of coarse δ-ferrite cannot effectively hinder microcrack propagation. Hence, cleavage morphology formed in δ-ferrite as microcrack propagating (Fig. 6). This was because {100} cleavage was a natural failure mode of many BCC crystals, including iron [65].

5. Conclusion

In the present study, the effect of annealing temperature on microstructure and mechanical properties of the low-density δ-TRIP steel of Fe-8.9Mn-4.6Al-0.29C via intercritical quenching-annealing was investigated. The austenitic stability and deformation behavior of the

presented steel was analyzed. Based on current experimental results, the main conclusions were as followed:

(1) The laminated hot-rolled experimental steel after intercritical quenching and annealing still maintained the micro-laminated microstructure, consisting of δ-ferrite band and tempered martensite lamellar. After annealing, a large number of fine austenite grains widely distributed in tempered martensite. With annealing temperature increasing, the proportion of austenite at room temperature increased. However, after annealing at 800 °C, some austenite underwent martensitic transformation during air cooling, which instead led to a decrease of austenite. At 620 °C of annealing temperature, carbides precipitated in tempered martensite region.
(2) The experimental steel after annealing treatment exhibited wide yield strength grades (460–620 MPa) under the conditions of extremely excellent ductility and low yield-to-tensile ratio, could substituted for TRIP 580T/400Y high strength automobile plates. Specifically, the 740 °C annealed steel had the highest product of strength and elongation, up to ~45.2 GPa%, and the maximum uniform elongation of ~62.4 %. The yield and tensile strength of annealed specimens decreased as annealing temperature, which attributed to the factors of the proportion of austenite, geometrically necessary dislocations density, and block boundary density of tempered martensite.
(3) The partitioning of C and Mn elements significantly affected the mechanical stability of austenite. Specifically, austenite in the specimen annealed at 680 °C possessed the highest stability. With annealing temperature increasing from 680 °C, austenitic stability decreased due to the dilution of austenitic alloy elements. When annealed at 620 °C, precipitated carbides in specimens led to the decrease of alloy elements in matrix, resulting in lower austenitic stability than that of 680 °C annealed specimens. Various austenitic stability contributed to various levels of TRIP effect during deformation.
(4) All annealed specimens exhibited three-stage work hardening characteristics, in which stage II and III resulted from the competition between dislocation gliding and TRIP effect, combined with the assistance of TWIP effect according to SFEs. The specimen annealed only at 680 °C exhibited apparently necking due to deficient TRIP effect, and afterward underwent a second stage of uniform elongation until failure.

CRediT authorship contribution statement

Feifan Li: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xueliang Shang:** Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jingxiao Zhao:** Software, Formal analysis. **Xuemin Wang:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Xiangyu Xu:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Zhongwen Wu:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influenced the work reported in this paper.

Data availability

Data will be made available on request.

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